

Antimicrobial Therapy in Veterinary Medicine

THIRD EDITION

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beta-lactamase is now widespread in *S. aureus*, particularly in clinical isolates, as a result of bacteriophage- or plasmid-mediated resistance. Among Gram-negative bacteria, plasmids encoding beta-lactamases have also become widespread and are the cause of extensive acquired resistance. Modification of PBPs is recognized to be increasingly important as another mechanism of resistance to penam penicillins. The most important example in human medicine is methicillin (oxacillin) resistance in *S. aureus* (MRSA) but resistance because of this mechanism has not been well documented in animal pathogens.

Pharmacokinetic Properties

The penicillins are organic acids that are generally available as the sodium or potassium salt of the free acid. In dry, crystalline form, penicillins are stable but lose their activity rapidly when dissolved. Apart from the isoxazolyl penicillins (cloxacillin, dicloxacillin, oxacillin) and penicillin V, acid hydrolysis limits the systemic availability of most penicillins from oral preparations. The penicillins (pK_a 2.7) are predominantly ionized in plasma, have relatively small apparent volumes of distribution (0.2–0.3 l/kg), and have short half-lives (0.5–1.2 hours) in all species of domestic animals. After absorption, they are widely distributed in the extracellular fluids of the body, but they cross biologic membranes poorly because they are ionized and poorly lipid soluble. Concentration in milk, for example, is about one-fifth that of serum. Entry across biologic membranes or through the blood-brain or blood-cerebrospinal fluid barrier is enhanced by inflammation, so that inhibitory drug concentrations may be attained at these sites, which are normally inaccessible to penicillin.

Penicillins are eliminated almost entirely by the kidneys, which results in very high levels in the urine; nafcillin is an exception, in that it is excreted mainly in bile. Renal excretion mechanisms include glomerular filtration and tubular secretion. The latter is subject to competitive inhibition by other organic acids, such as probenecid. Impaired renal function delays excretion of the penicillins, but the wide margin of safety of this class of drug offsets the absolute need to adjust dosage.

Drug Interactions

Penicillins are usually synergistic with the aminoglycosides against many bacteria, which are susceptible to each drug alone, because they enhance penetration of the aminoglycoside. Such synergism may occur with penicillinase-producing *S. aureus*. Penicillins are synergistic against these organisms with drugs that bind beta-lactamase enzymes, such as cloxacillin, clavulanic acid, sulbactam, tazobactam, and some cephalosporins. Aminobenzylpenicillins and ureidopenicillins are increasingly combined with beta-lactamase inhibitors (see Chapter 8).

Toxicity and Adverse Effects

Penicillins and beta-lactam antibiotics generally are remarkably free of toxic effects even at doses grossly in excess of those recommended. The major adverse effects are acute anaphylaxis and collapse; milder hypersensitivity reactions (urticaria, fever, angioneurotic edema) are more common. All penicillins are cross-sensitizing and cross-reacting, but cross-reactions occur in only about 5–8% of human patients treated with cephalosporins. Anaphylactic reactions are less common after oral administration than after parenteral administration. Penicillins must not be used in animals known to be sensitive. Less common adverse reactions include hemolytic anemia and thrombocytopenia.

Dosage Considerations

Beta-lactams produce killing and lysis of bacteria at concentrations above MIC. Post-antibiotic effects are observed only for staphylococci in vivo, so that dosage requires that drug concentrations exceed MIC for most of the dosage interval. Excessive drug concentrations may be counterproductive because of the Eagle effect described earlier, in which sometimes dramatic reduction of killing occurs in the presence of high, supra-MIC concentrations of the beta-lactam antibiotic.

Clinical Usage

Penicillins are important antibacterial drugs in the treatment of infections in animals (see Table 6.1). The often exquisite susceptibility of Gram-positive bacteria, such as the beta-hemolytic streptococci, means that benzyl penicillin is often a drug of choice for these infections, because of its high potency and low toxicity. Antistaphylococcal penicillins are in widespread use in the prevention and treatment of staphylococcal mastitis in cows. The extended-spectrum penicillins, particularly aminobenzylpenicillins, have lost much of their potency against Gram-negative bacteria over the decades but have been revitalized by their combination with beta-lactamase inhibitors (see Chapter 8). The antipseudomonal penicillins remain important for this activity but are rivaled by antipseudomonal cephalosporins.

Group 1, Benzyl Penicillin and Long-acting Parenteral Forms

Sodium benzyl penicillin G is available as the benzyl, the procaine benzyl, and the tribenzyl ethylenediamine (benzathine) forms. Frequent dosing of benzyl penicillin is required because of the drug's rapid excretion, so that long-acting, delayed-absorption forms (procaine, benzathine) have been developed, with procaine penicillin being the most extensively used because dosing frequency is usually q24 hours. The principle behind the

use of procaine and benzathine penicillin is that both forms delay absorption from the injection site. Thus, although the elimination half-life is the same, the absorption half-life is much longer, reducing the need for frequent dosing. Delayed absorption also means a lower peak concentration.

Antimicrobial Activity

The activity of penicillin G was originally defined in units. Crystalline sodium penicillin G contains approximately 1,600 units/mg (1 unit = 0.6 µg; 1 million units of penicillin = 600 mg, or 0.6 g). Most semisynthetic penicillins are prescribed by weight (mg/kg) rather than units.

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Good Susceptibility (MIC ≤ 0.12 µg/ml) is shown by many aerobic Gram-positive bacteria, including all beta-hemolytic streptococci (such as *Streptococcus agalactiae*, *S. canis*, *S. zooepidemicus*, *S. dysgalactiae*, *S. suis*, *S. uberis*, *Bacillus anthracis*, *Actinomyces* spp., most corynebacteria (including *C. pseudotuberculosis*, *C. renale*), *Erysipelothrix rhusiopathiae*, and most *Listeria monocytogenes* (Table 6.2). Susceptible anaerobes include *Clostridium* spp., most *Fusobacterium* spp., and some *Bacteroides*. Susceptible Gram-negative aerobes include *Haemophilus somnus*.

Variable Susceptibility is shown by *Staphylococcus aureus* and other staphylococci, although in the absence of resistance, staphylococci are highly susceptible.

Moderate Susceptibility (MIC 0.25–2 µg/ml), which may sometimes vary because of acquired resistance, is shown by *Actinobacillus* spp., *Borrelia* spp., *Brucella* spp., *Haemophilus* spp., *Leptospira* spp., *Moraxella* spp., *Pasteurella* spp., *Proteus* spp., *Taylorella equigenitalis*, and *Serpulina* spp.

Resistance (MIC ≥ 4 µg/ml) is shown by Enterobacteriaceae (other than some *Proteus* spp.), *Bacteroides fragilis*, *Bordetella* spp., most *Campylobacter* spp., and *Nocardia* spp.

Antibiotic Resistance

Despite extensive use of penicillin in veterinary medicine for many years, most Gram-positive bacteria remain susceptible to the drug. *Staphylococcus aureus* is an exception. The beta-lactamase enzymes of *S. aureus* are mainly active against penicillin G, ampicillin, and carbenicillin but hydrolyze penicillinase-stable penicillins (methicillin, cloxacillin) and cephalosporins poorly. Resistance in usually susceptible Gram-negative bacteria such as *Haemophilus* and *Pasteurella* is the result of R plasmid-mediated production of beta-lactamases, but this is rarer.

Pharmacokinetic Properties

Pharmacokinetic properties of the benzyl penicillins were discussed earlier in this chapter under Penam Penicillins: Pharmacokinetic Properties. Acid hydrolysis in the stomach limits the systemic availability of benzyl penicillin administered orally.

Drug Interactions

Penicillin G is synergistic with the aminoglycosides against many Gram-positive bacteria, except those showing high-level aminoglycoside resistance. Such synergism may be seen even with penicillinase-producing *Staphylococcus aureus*. Penicillin is synergistic against these organisms with drugs that bind beta-lactamase enzymes (see Chapter 8). Penicillin G has been combined with streptomycin for use in animals, but there is little clinical evidence supporting the clinical value of the combination (Whittem and Hanlon, 1997). For this reason, because streptomycin is associated with tissue residues, the combination is no longer available in some countries, including the United States. In addition, there are significant differences in pharmacokinetic properties among different combined preparations (Groen et al., 1996).

Toxicities and Adverse Effects

The parent benzyl penicillin and its numerous derivatives are relatively safe drugs; toxic effects were described under Penam Penicillins: General Considerations. Many of the acute toxicities reported in animals are the result of the toxic effects of the potassium or procaine with which penicillin is combined in the dosage form. To avoid cardiac arrest, care should be taken with the rate at which potassium penicillin G is injected IV; administration of the sodium salt is safer. Procaine penicillin G should never be given IV. In high doses given IM, the procaine form may cause nervous excitement (incoordination, ataxia, excitability) and death, particularly in horses (Nielsen et al., 1988). It should not be administered to horses within 2 weeks before a race, so as to avoid procaine-positive drug test results. Procaine penicillin should be stored in the refrigerator and not used past expiration dates; repeated use of the same injection site should be avoided, especially in horses. Severe, immune-mediated hemolytic anemia with icterus has been reported in horses (Blue et al., 1987).

Administration and Dosage

Recommended dosages for penam penicillins are shown in Table 6.3. Because of the relative lack of toxicity of penicillins, their dosage can be tailored, to some extent, to the susceptibility of the infecting bacteria more than with any other class of antibiotic. The effectiveness of penicillin therapy is related to the time that tissue concentration

Ziv G, et al. 1983. Concentrations of methicillin in blood, normal milk, and mastitic milk of cows after intramuscular injection of methicillin and amethicillin. J Vet Pharm Ther 6:41.

Group 4, Extended-spectrum Penicillins: Aminobenzylpenicillins

AS.1

Aminobenzylpenicillins (ampicillin, amoxicillin, and the related esters bacampicillin, hetacillin, pivampicillin, and talampicillin) have similar antimicrobial activity, but amoxicillin and possibly pivampicillin have the advantage of achieving higher tissue concentrations because of better absorption from the intestine. The broad-spectrum aminobenzylpenicillins are slightly less active than penicillin G against Gram-positive and anaerobic bacteria and also are susceptible to staphylococcal penicillinase. These broad-spectrum drugs have considerably greater activity against Gram-negative bacteria such as *E. coli*, *Proteus mirabilis*, and *Salmonella*. Nevertheless, acquired resistance has considerably reduced the effectiveness of these drugs. An exciting development has been their combination with beta-lactamase-inhibiting drugs, which increases their effectiveness considerably (see Chapter 8) and with which these drugs should generally be combined.

Antimicrobial Activity

see benzyl penicillin

Good Susceptibility (MIC ≤ 1): As for benzyl penicillin group but includes *Borrelia* spp. and *Leptospira* spp., which are highly susceptible, as well as *Actinobacillus* spp., *Haemophilus* spp., *Moraxella* spp., and *Pasteurella* spp. (Tables 6.2 and 6.5).

Moderate Susceptibility (MIC 2–4 $\mu\text{g/ml}$): As for benzyl penicillin but also *Campylobacter* spp., enterococci, and *Rhodococcus equi*. Variable moderate activity (because of acquired resistance) against *E. coli*, *Proteus mirabilis*, and *Salmonella*. Acquired resistance in Enterobacteriaceae is widespread.

Resistance (MIC $>4 \mu\text{g/ml}$, approximately): *Bacteroides fragilis*, *Bordetella bronchiseptica*, *Citrobacter* spp., *Enterobacter* spp., *Klebsiella* spp., other *Proteus* spp., *Pseudomonas aeruginosa*, *Serratia* spp., and *Yersinia enterocolitica*.

Antimicrobial Resistance

Plasmid-mediated acquired resistance is common in Gram-negative bacteria and is often multiple, such as that in most enterotoxigenic *E. coli* and *Salmonella typhimurium*. Many *E. coli* that cause bovine mastitis are resistant. Aminobenzylpenicillins are susceptible to *Staphylococcus aureus* beta-lactamase (see Tables 6.2 and 6.5).

TABLE 6.5. In vitro activity ($\mu\text{g/ml}$) of new penicillins against various medically important opportunist bacteria

Organism	Ampicillin		Mecillinam		Ticarcillin		Azlocillin		Piperacillin	
	MIC ₅₀	MIC ₉₀	MIC ₅₀	MIC ₉₀	MIC ₅₀	MIC ₉₀	MIC ₅₀	MIC ₉₀	MIC ₅₀	MIC ₉₀
<i>Streptococcus agalactiae</i>	0.06	0.12	2	8	2	4	0.25	1	0.25	1
<i>Escherichia coli</i>	4	128	1	4	16	128	8	128	8	128
<i>Klebsiella pneumoniae</i>	128	128	2	128	128	128	32	128	8	128
<i>Citrobacter diversus</i>	4	128	0.5	4	16	128	4	8	4	4
<i>Enterobacter cloacae</i>	128	128	2	32	8	128	4	32	4	32
<i>Proteus mirabilis</i>	1	4	4	16	0.5	16	0.5	16	0.5	16
<i>Pseudomonas aeruginosa</i>	128	128	128	128	16	128	4	128	4	128
<i>Bacteroides</i> spp. ^a	1	32	2	16	4	32	2	8	2	4

Source: Modified from Prince and Neu (1983) with permission.

^aOther than *B. fragilis*.